Comparison of Numerical Integrators for N-Body Simulation of the Solar System: Euler, Velocity Verlet, RK4 and Adaptive RK45

Wabi Tafese, Rakeb Tewodros, Henok Yared, Rediet Mebt, Aklesiya Negalign, Hemen Tadese, Arsema Laikemariam

Independent student researchers, Ethiopia · Corresponding author: Wabi Tafese, wabitafese9@gmail.com

Project work: February 2026 · This version: October 2026 · DOI: https://doi.org/10.5281/zenodo.23529877

Keywords: N-body simulation, Runge-Kutta, Verlet integration, orbital mechanics, numerical integration, energy conservation, solar system

Abstract

Simulating planetary motion requires numerical integration of the gravitational N-body problem, which has no closed-form solution for more than two bodies. We built an interactive, browser-based three-dimensional solar-system simulator that runs four integrators in parallel from identical initial conditions: explicit Euler, velocity Verlet, the classical fourth-order Runge-Kutta method (RK4), and the adaptive Dormand-Prince RK45 method. Initial conditions come from JPL's approximate Keplerian elements at epoch J2000, and calculations use astronomical units, years and solar masses. We compare the methods by relative energy error, angular-momentum error and position error against a high-accuracy reference solution. Over 100 simulated years with a 1.83-day step, explicit Euler loses 18% of its energy and its orbits spiral outward, velocity Verlet keeps energy error near 3.5×10^{-7} and conserves angular momentum to machine precision, fixed-step RK4 reaches 1.7×10^{-5} with a slow upward drift, and adaptive RK45 reaches 6.8×10^{-12}. RK4 shows approximately fourth-order convergence in position. The results show that no single method wins on every measure: Verlet is the more robust choice for long, coarse-step runs, while RK4 and RK45 are more accurate at fine steps. The software, tests and data are open source.

1. Introduction

Planetary motion has driven astronomy, physics and space exploration for centuries. With two bodies, Newton's law of gravitation yields Kepler's elliptical orbits exactly. With three or more bodies there is no general closed-form solution, so predictions depend on numerical integration [Murray and Dermott, 1999]. The choice of integrator decides whether a simulated orbit stays stable or slowly drifts away from physical reality, so it matters for both education and research.

This work has four objectives:

1. Model gravitational interactions between the Sun and the eight planets in three dimensions using real orbital data.
2. Implement four integrators and run them side by side from identical initial conditions.
3. Measure their accuracy, conservation of energy and angular momentum, and convergence order.
4. Provide an accessible, tested, open-source tool that lets students see these differences directly.

Contributions. (i) A reproducible comparison of four integrators on the real solar-system initial conditions; (ii) an interactive web application with live energy-error diagnostics; (iii) an automated

test suite and benchmark scripts that regenerate every number and figure in this paper.

2. Background and Related Work

2.1 Newtonian gravity

Two bodies with masses m_1 and m_2 separated by distance r attract with force $F = G\, m_1\, m_2 / r^2$, where G is the gravitational constant. For N bodies, the acceleration of body i is

$$a_i = G \sum_{j \neq i} m_j \, (r_j - r_i) / |r_j - r_i|^3 .$$

Because the force on i from j is the negative of the force on j from i (Newton's third law), each pair is evaluated once, giving $O(N^2/2)$ work per acceleration evaluation.

2.2 Numerical integrators

Explicit Euler is first order and is known to add energy to orbital systems. Velocity Verlet [Verlet, 1967] is second order, time-reversible and symplectic, so its energy error stays bounded for Hamiltonian systems [Hairer et al., 2006]. The classical Runge-Kutta method (RK4) is fourth order but not symplectic, so its energy error can drift slowly over very long runs. Embedded pairs such as Dormand-Prince RK45 [Dormand and Prince, 1980] estimate local error and adapt the step size [Press et al., 2007]. Production N-body codes such as REBOUND [Rein and Liu, 2012] use symplectic and high-order schemes for the same reasons.

2.3 Gap addressed

Many educational simulators use a single method with an arbitrary time step and unitless constants. Here we use physical units, real initial conditions and several methods at once, and we quantify the differences instead of describing them qualitatively.

3. Methods

3.1 Units and equations of motion

We use astronomical units (AU), years and solar masses ($M\odot$). In these units $G = 4\pi^2\ \mathrm{AU}^3\ M\odot^{-1}\ \mathrm{yr}^{-2}$, so a circular orbit at 1 AU has a period of exactly 1 year. The state of each body is its position r and velocity v, and the system evolves by $dr/dt = v$, $dv/dt = a(r)$ with a given in Section 2.1. No softening is applied.

3.2 Initial conditions

For each planet we use the JPL approximate Keplerian elements at J2000 [Standish and Williams, JPL]: semi-major axis a, eccentricity e, inclination I, mean longitude L, longitude of perihelion ϖ and longitude of ascending node Ω. We set the argument of perihelion $\omega = \varpi - \Omega$ and the mean anomaly $M = L - \varpi$, solve Kepler's equation $M = E - e \sin E$ by Newton iteration, and compute the position and velocity in the orbital plane:

$$x = a(\cos E - e),\ y = a\sqrt{1-e^2}\,\sin E,\ \dot{x} = -\sqrt{Ga}\,\sin E / r,\ \dot{y} = \sqrt{Ga(1-e^2)}\,\cos E / r,\ r = a(1 - e \cos E).$$

These are rotated through ω, I and Ω into the ecliptic frame. All positions and velocities are then shifted so the system's centre of mass is at rest at the origin. Planetary masses are the standard values in solar masses (for example Earth 3.003×10^{-6} and Jupiter 9.55×10^{-4}). The elements are mean elements applied as osculating values at J2000, so orbits differ slightly from the real ephemeris (Section 6).

3.3 Integrators

With state (x, v), step h and acceleration a(x):

Explicit Euler: $x \leftarrow x + h\,v$; $v \leftarrow v + h\,a(x)$.

Velocity Verlet: $x \leftarrow x + h\,v + \tfrac{1}{2}h^2\,a$; compute $a' = a(x_{new})$; $v \leftarrow v + \tfrac{1}{2}h\,(a + a')$.

RK4: $k_1 = f(y)$, $k_2 = f(y + \tfrac{1}{2}h\,k_1)$, $k_3 = f(y + \tfrac{1}{2}h\,k_2)$, $k_4 = f(y + h\,k_3)$; $y \leftarrow y + (h/6)(k_1 + 2k_2 + 2k_3 + k_4)$, where $y = (x, v)$ and $f(y) = (v, a(x))$.

Adaptive RK45 (Dormand-Prince): a seven-stage embedded fourth/fifth-order pair. A step is accepted when the scaled error is at most 1 (relative tolerance 10^{-11}, absolute tolerance 10^{-13}), and the next step is $h \leftarrow h \cdot \min(5, \max(0.2, 0.9 \cdot \mathrm{err}^{-1/5}))$, never exceeding the nominal step.

3.4 Diagnostics

Total energy is $E = \Sigma\, \tfrac{1}{2}\, m_i\, v_i^2 - \Sigma_{i<j}\, G\, m_i\, m_j / r_{ij}$, and total angular momentum is $L = |\Sigma\, m_i\, r_i \times v_i|$. We report the relative errors $|\Delta E/E_0|$ and $|\Delta L/L_0|$. Position error is the distance between Earth's position and its position in a reference run (RK45, energy error about 10^{-12} over 10 years) at the same time.

3.5 Software implementation

The simulator is written in JavaScript with three.js for 3D rendering and runs entirely in the browser; the physics modules have no dependencies and also run in Node.js. The source is organised into modules for body data, gravity, orbital-element conversion, integrators, massless test particles and rendering. The physics step is fixed and independent of the display frame rate. The application also includes an asteroid belt of 4,000 massless particles and a comet launcher, which move under the Sun, Jupiter and Saturn with a leapfrog scheme. Moons are drawn for illustration only and are not part of the integrated physics.

3.6 Experimental setup

Two experiments were run with the same code that the web application uses.

1. **Convergence (10 years):** steps of 14.6, 7.3, 3.65, 1.83, 0.91 and 0.46 days for Euler, Verlet and RK4, with errors measured at t = 10 years.
2. **Long-run drift (100 years):** all four methods at a step of 1.83 days (0.005 yr), with energy error sampled once per simulated year.

All numbers in this paper are produced by *npm run bench* and *npm test* in the project repository.

4. Results

4.1 Long-run energy conservation

Table 1 gives the relative energy error after 100 years at a step of 1.83 days.

Method	Order	Relative energy error after 100 yr
Explicit Euler	1	1.8×10^{-1}
Velocity Verlet	2	3.5×10^{-7}
RK4	4	1.7×10^{-5}
Adaptive RK45	4(5)	6.8×10^{-12}

Table 1. Energy error after 100 simulated years, step 1.83 days.

Euler's error grows steadily and its orbits spiral outward: in an earlier 100-year test Earth's distance from the Sun increased to about 5.6 AU. Verlet's error oscillates and does not grow, and its total angular momentum is conserved to about 10^{-15}. RK4's error starts near 10^{-7} and increases steadily to 1.7×10^{-5} at year 100, as expected for a non-symplectic method, while Verlet's oscillates, mostly between about 10^{-8} and 10^{-5} (with one brief dip to roughly 10^{-10} near year 45), without growing. At this step the two are of similar size, but only Verlet's error is bounded.

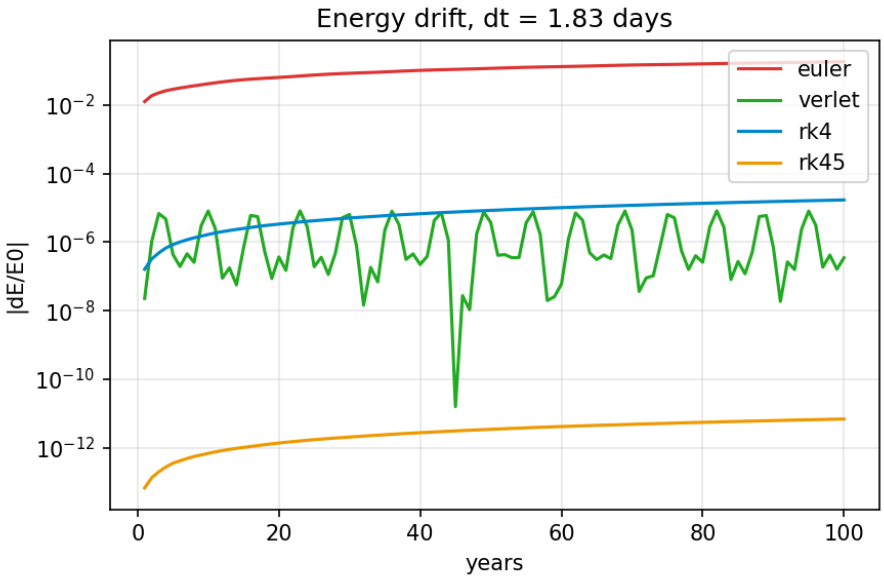


Figure 1. Relative energy error $|\Delta E/E_0|$ over 100 simulated years for the four integrators (step 1.83 days, logarithmic vertical axis).

4.2 Convergence

Step (days)	Euler: energy / Earth position error (AU)	Verlet: energy / position error (AU)	RK4: energy / position error (AU)
14.61	1.6×10^{-1} / 1.32	1.8×10^{-6} / 1.24	2.1×10^{-2} / 9.1×10^{-2}
7.31	1.0×10^{-1} / 2.53	5.0×10^{-5} / 0.34	2.9×10^{-2} / 3.0×10^{-3}
3.65	6.4×10^{-2} / 2.77	2.8×10^{-5} / 8.5×10^{-2}	5.6×10^{-5} / 1.0×10^{-4}
1.83	4.2×10^{-2} / 2.96	8.1×10^{-6} / 2.1×10^{-2}	1.7×10^{-6} / 3.9×10^{-6}
0.91	2.9×10^{-2} / 2.95	2.1×10^{-6} / 5.3×10^{-3}	5.2×10^{-8} / 1.7×10^{-7}
0.46	2.0×10^{-2} / 2.23	5.3×10^{-7} / 1.3×10^{-3}	1.6×10^{-9} / 7.8×10^{-9}

Table 2. Errors after 10 simulated years versus step size.

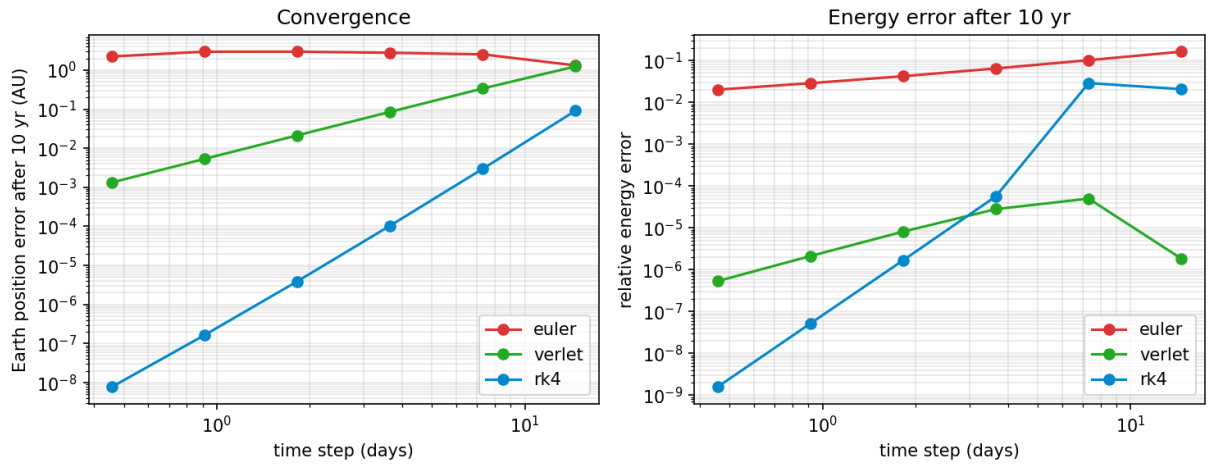


Figure 2. Left: Earth position error after 10 years versus step size. Right: relative energy error after 10 years versus step size. Both axes are logarithmic.

The observed orders match theory. Verlet's position error falls by a factor of about 4 each time the step is halved (second order). RK4's falls by a factor of 21 to 27 over the finer steps, consistent with at least fourth order (the factor is 16 for exactly fourth order; the measured value is somewhat higher in this range). Euler's energy error falls by only about 1.5 per halving (first order), and its position error is so large that Earth is effectively in the wrong place after 10 years at every step tested. At the two coarsest steps (14.6 and 7.3 days) the 88-day orbit of Mercury is covered by only 6 to 12 steps, and RK4's energy error is accordingly poor there (2 to 3 percent), while Verlet's is not; coarse steps are where the symplectic method has the clearest advantage. Verlet's energy error is not monotonic in the step size (for example, 1.8×10^{-6} at 14.6 days but 5.0×10^{-5} at 7.3 days). This is expected for a symplectic method: its energy error oscillates rather than accumulating, so its value at a fixed end time depends on the phase of that oscillation at t = 10 years, and the position error is the cleaner measure of convergence.

4.3 Adaptive stepping

Adaptive RK45 reached an energy error of 7×10^{-13} over 10 years and 6.8×10^{-12} over 100 years, about six orders of magnitude better than fixed-step RK4 at the same nominal step, at the cost of several force evaluations per accepted step (seven stages, with failed steps repeated). Because RK45 chooses its own step size, never larger than the nominal one, this is a comparison of accuracy at equal nominal step rather than at equal computational cost.

4.4 Validation

The automated suite of eight tests passes. Among other checks: the total force on the system sums to zero (Newton's third law); the barycentre starts at rest at the origin; initial semi-major axes computed from the state vectors agree with the JPL table to within 1%; RK4's error shrinks by a factor between 10 and 24 when the step is halved; Verlet conserves angular momentum to 10^{-10} over 50 years; Euler's energy error is more than 100 times Verlet's; and a massless test particle started on a circular 1 AU orbit stays within 1% of its radius for 5 years. In the in-browser self-test, Earth's measured orbital period is 1.002 years against the expected 1.000 (within a 0.4% tolerance).

4.5 Performance

The cost of one acceleration evaluation scales as $O(N^2/2)$. With $N = 9$ bodies all four methods run many steps per displayed frame in a standard browser, so the simulation runs in real time. We did not conduct a controlled runtime benchmark; measured wall-clock times were dominated by timer noise and JIT warm-up and are therefore not reported.

5. Discussion

The four methods answer different needs. For short, accurate runs at fine steps, RK4 and especially adaptive RK45 give the smallest errors. For long runs at coarse steps, as in a real-time visualisation, symplectic Verlet is the safer default because its energy error stays bounded and its angular momentum is conserved exactly. Explicit Euler is unsuitable for orbital mechanics at any step we tested.

Our results correct a common simplification: RK4 is not simply "more stable" than Verlet. It has higher order and smaller error per step, but it lacks the structure-preserving property, and its energy error drifts upward over time while Verlet's does not. Choosing between them depends on step size and run length. Accurate orbits require balancing step resolution against computation, and the step must resolve the fastest orbit in the system (here Mercury).

Earlier versions of this project stated that multi-body systems "demonstrate chaotic behaviour". We have removed that claim, because we did not measure chaos (for example with Lyapunov exponents) and the solar system's planetary orbits are regular on the time scales studied.

6. Limitations

The JPL elements are approximate mean elements used as osculating values; they are not full JPL Horizons state vectors, so positions differ slightly from the real ephemeris. Only the Sun and eight planets are integrated; there are no moons, asteroids or dwarf planets in the main N-body system. The belt and comets are massless and feel only the Sun, Jupiter and Saturn. No relativistic effects are modelled, so Mercury's perihelion precession is absent. Collisions are limited to comets being absorbed by the Sun; planets do not collide or merge.

Position errors are measured against our own high-accuracy RK45 run, not against an external ephemeris. A comparison with JPL Horizons is future work. Runtime was not benchmarked. The

study uses one set of initial conditions (J2000); results may vary for other epochs.

7. Future Work

1. Validate positions against JPL Horizons state vectors.
2. Add higher-order symplectic integrators (for example Yoshida or Wisdom-Holman methods) and compare them with Verlet and RK45.
3. Move the physics to a Web Worker and add Barnes-Hut or GPU acceleration for large particle counts.
4. Add collision detection with merging or bouncing, and moons in the N-body system.
5. Measure chaos with Lyapunov exponents and sensitivity to initial conditions.
6. Run a controlled performance benchmark across devices.

8. Conclusion

We compared four integrators on the real solar-system N-body problem using open, reproducible code. Verlet provides bounded energy error and exact angular-momentum conservation at coarse steps, RK4 gives fourth-order convergence and the best accuracy per step at fine steps, adaptive RK45 gives the lowest overall error, and explicit Euler fails. The accompanying simulator makes these differences visible in real time and offers a tested foundation for education and future extensions.

Code and Data Availability

Source code, tests, benchmark scripts and result data are available at https://github.com/wabix2/realistic-solar-system (folder rk4-labs). A live demonstration is at https://rk4-labs.vercel.app. A permanent archived version of this paper is available at Zenodo: https://doi.org/10.5281/zenodo.23529877. The software is released under the MIT licence.

Author Contributions

Wabi Tafese: Lead Developer and Researcher; Rakeb Tewodros: Programmer; Henok Yared: Editor; Rediet Mebt: Presentation; Aklesiya Negalign: Data Analyst; Hemen Tadese: Support; Arsema Laikemariam: Coordinator.

References

1. Dormand, J. R., and Prince, P. J. (1980). A family of embedded Runge-Kutta formulae. *Journal of Computational and Applied Mathematics*, 6(1), 19–26.
2. Goldstein, H., Poole, C., and Safko, J. (2002). *Classical Mechanics* (3rd ed.). Addison-Wesley.
3. Hairer, E., Lubich, C., and Wanner, G. (2006). *Geometric Numerical Integration* (2nd ed.). Springer.
4. Murray, C. D., and Dermott, S. F. (1999). *Solar System Dynamics*. Cambridge University Press.
5. Newton, I. (1687). *Philosophiae Naturalis Principia Mathematica*. London.
6. Press, W. H., Teukolsky, S. A., Vetterling, W. T., and Flannery, B. P. (2007). *Numerical Recipes: The Art of Scientific Computing* (3rd ed.). Cambridge University Press.
7. Rein, H., and Liu, S.-F. (2012). REBOUND: an open-source multi-purpose N-body code for collisional dynamics. *Astronomy and Astrophysics*, 537, A128.
8. Standish, E. M., and Williams, J. G. Keplerian Elements for Approximate Positions of the Major Planets. NASA Jet Propulsion Laboratory, Solar System Dynamics (ssd.jpl.nasa.gov).

9. Verlet, L. (1967). Computer “experiments” on classical fluids. I. Thermodynamical properties of Lennard-Jones molecules. *Physical Review*, 159(1), 98–103.